Ramification Index v3 — Self-Audit (Full)

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Contents

v3 methods — the ag-ram-index harness, ported and run

Status: harness built, battery run, results below. Every figure here traces to `analysis/out/FACTS.txt`, which is verbatim run output. That convention is inherited from `ag-ram-index/papers/FACTS.txt` and exists so a number cannot drift between the code and the prose.

```
.venv/bin/python analysis/battery.py > analysis/out/FACTS.txt
```

```
cd analysis && ../.venv/bin/python -m pytest -q
```

Source of the method: `~/claude-code/ag-ram-index`, which tested six candidate agricultural leading indicators, found none, and produced **four results that looked publishable and were wrong**. Each survived every check in place at the time and died only when a further guard was added. Those guards are now in `analysis/ri_leadtest/leadtest.py`, pinned by 20 regression tests.

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1. Headline

Nothing in the public semiconductor price complex leads the macroeconomy once the guards are applied, and the published Ramification Index panel is too small to tell either way. Those are different findings and the difference is the whole point of §4.

The one thing that survived every guard in the agricultural study was a mirage created by the data vendor. The equivalent check here (§3) says the public monthly proxy is **not measuring DRAM** — so the monthly track, which is the thing you remembered wanting, is currently blocked on data acquisition rather than on method.

2. On the monthly question

I could not recover the original remark. The `ag-ram-index` session transcripts are not on this machine (`~/.claude/projects/` has no entry for that directory), and a full-text search across every other local transcript returns nothing. So the following is reconstruction from the artifacts, not recall.

The case for monthly is strong and the ag repo makes it implicitly everywhere: **every substantive test in that study ran on monthly data** — 554 months of potash, 792 of Pink Sheet prices, 826 of heifer slaughter, 992 of fertilizer. The harness is built for that frequency. Three of its guards are barely estimable without it:

- point-in-time orthogonalization has a **120-observation burn-in** before it emits a single residual;
- the contemporaneous control fits `2k+1` regressors and needs residual degrees of freedom to spare;
- a 12-lag Bonferroni sweep at $\alpha = 0.00417$ needs real power behind it.

The published RI panel is **annual, n = 46**. That is not enough for any of them.

What monthly is actually worth — the power calculation

Minimum partial correlation detectable at 80% power against the Bonferroni-corrected α, by panel:

Panel	n	lags	α	min detectable partial *r*
Published annual panel	46	3	0.0167	**0.508**
v2 quarterly window 1985–2024	160	8	0.00625	0.360
Full quarterly	237	8	0.00625	0.299
Monthly	712	12	0.00417	**0.192**

The annual panel cannot see anything smaller than a partial correlation of **0.51**. That is an enormous macroeconomic effect — larger than almost any published leading indicator. Monthly drops the floor to **0.19**.

This reframes the annual result completely. See §4.

The blocker

There is **no free monthly DRAM ASP series.** TrendForce / DRAMeXchange is commercial. And the McCallum archive cited in the v2 paper — `jcmit.net/memoryprice.htm` — **no longer resolves to the archive**; the domain now serves unrelated commercial content, and the Wayback copy is behind a challenge page. A paper whose primary 1980–2017 source has gone dark has a reproducibility problem independent of any of this.

So the monthly battery had to run on the closest public monthly price series, `PCU334413334413` (PPI: Semiconductor & Related Device Manufacturing, 713 months, 1967–2026) — and then Guard 7 was run to find out whether that was legitimate. It was not. See §3.

3. Guard 7 killed the substitution — exactly as it did in agriculture

The last survivor of the agricultural study died on the discovery that two crop series from independent providers correlate **0.794** while the two fertilizer series correlate **0.175**: they were not measuring the same good, and the lead followed the vendor rather than the economy.

Run here, on annual log-differences over the common 1981–2026 span:

Pair	n	r	same good?
Public semi PPI vs the paper's own DRAM ASP	46	**0.186**	**no**
semi PPI vs broader component-industry PPI	498	0.655	yes
semi PPI vs electronic-components commodity PPI	712	0.519	no
semi PPI vs its own product-line index	599	0.286	no
semi PPI vs storage-device PPI	402	−0.024	no

0.186. Effectively the fertilizer number. The public semiconductor PPI and the Ramification Index do not move together, so every monthly result in §5 is a statement about the semiconductor price complex and **not** about DRAM ASP. It can neither confirm nor refute v2.

Two further readings worth having:

- Even *within* BLS, two constructions of semiconductor prices correlate 0.286. “The semiconductor price” is not a well-defined single object. Any v3 claim about DRAM must name its series and show the substitution test.
- v2’s existing three-vendor check (Gartner / TrendForce / Counterpoint, r = 0.38–0.44 against GDP) compares *correlations with GDP*, not the ASP series with each other. Those are different tests and only the second one is Guard 7. **The vendors’ ASP series need to be correlated against each other.**

4. The published index: underpowered, not refuted

The only track that uses the actual Ramification Index as published:

```
ram_index_R -> real GDP growth, annual, n = 46, 3 lags
  best forward:  lag 2, F = 4.09, p = 0.0243
  reverse:       lag 2,           p = 0.499    (F ratio 5.78)
  Bonferroni α = 0.0167
  VERDICT: NO LEAD — p = 0.0243 fails α = 0.0167
```

Read the components rather than the verdict:

- **Guard 2/3 pass cleanly.** Reverse p = 0.499, forward/reverse F ratio 5.78, well above the 2.0 dominance threshold. The direction is unambiguous.
- **Guard 5 passes** — and this is the guard v2 has never run and the one I expected to do the damage. Adding the contemporaneous term retains **98.9%** of the F statistic (4.094 $\rightarrow$ 4.050, p = 0.025). Whatever this is, it is anticipation, not persistence proxying for same-period co-movement. Contrast the ag study's heifer result, which retained so little it died outright.
- **Guard 1 fails, narrowly.** p = 0.0243 against α = 0.0167.
- Dropping the forecast row (§6) barely moves it: p = 0.0330, 99.9% F retained.

Now combine that with the power table: this panel could only have detected a partial correlation of 0.51 or larger. **A test with no power cannot reject anything.** The correct label for the published annual panel is not NO LEAD; it is *underpowered*, and the ag study has a name for that class of outcome — UNTESTABLE, the verdict it gave potash prices rather than reporting a null.

This is the strongest argument for monthly, and it is a constructive one: the RI is not failing the guards. It is failing to have enough observations to face them.

5. What the monthly battery found (about semiconductors, not DRAM)

Caveat from §3 applies to everything here.

Test	n	Verdict
semi PPI $\rightarrow$ INDPRO	712	NO LEAD (p = 0.0122 vs α = 0.00417)
semi PPI $\rightarrow$ PAYEMS	712	NO LEAD (p = 0.329)
semi PPI ex-PPIACO, point-in-time $\rightarrow$ INDPRO	592	NO LEAD (p = 0.0163)
semi PPI $\rightarrow$ real GDP, quarterly	237	NO LEAD (p = 0.046)
semi PPI $\rightarrow$ real GDP, **v2's own 1985–2024 window**	160	NO LEAD (p = 0.155)

And every one of them **dies under Guard 5** — the lag block never survives the contemporaneous term. Monthly signal autocorrelation is only ρ_1 = 0.161, so this is not a persistence artifact; there is simply nothing there.

The YoY contrast

`semi PPI` $\rightarrow$ `INDPRO`: log-diff p = 0.0122, YoY p = 0.0345. `semi PPI` $\rightarrow$ `PAYEMS`: the forward/reverse F ratio swings from **140.2 under log-differences to 0.27 under YoY** — the transform inverts the apparent direction of the relationship. Neither verdict changes here, but the machinery that flipped the ag study's conclusions is demonstrably live in this data. A synthetic test pinning the effect is in the test suite (`test_yoy_transform_manufactures_a_lead_that_logdiff_denies`).

The placebos, and a problem with v2's

Capacitor and resistor PPIs — commodity passives, bought short-lead against realised orders — return clean nulls (p = 0.443, p = 0.328). Good.

The storage-device PPI leads INDPRO at 8 months (p = 4.4e-04, F ratio 7.89) and is the only price series in the entire run that survives Guard 5 (85.1% F retained, controlled p = 0.030).

That is a problem for v2's robustness section, in two ways. Empirically, v2 reports NAND as a *failing* placebo (3/6 recessions vs 6/6); here the storage analogue outperforms the semiconductor signal. Structurally, it was never a valid placebo to begin with: v2's own mechanism is that DRAM leads because a few hundred capacity-planning firms buy it ahead of deploying compute. NAND is bought by *the same firms, for the same deployments, on the same planning horizon*. On v2's own argument NAND should lead too. A placebo has to fail the mechanism, not merely be a different product.

6. Guard 4 audit of the published panel

The last row is a forecast. `data/indices-wide.csv` is annual and ends at 2026, constructed part-way through 2026:

	ram_index_R	hbm_weight	real_gdp_yoy	unemp_u3	sp500_close
2024	0.477	0.18	2.8	4.0	5881.63
2025	0.346	0.30	2.2	4.1	6089.54
2026	**0.863**	**0.42**	**1.5**	**4.4**	**5995**

Every 2026 value is a projection standing in a column of realised history. Guard 4 says a value at t must not depend on information unavailable at t; a forecast is the limiting case. The panel should carry a vintage column, or the 2026 row should be marked as provisional. The estimates are reported both ways in FACTS.txt and the difference is small — but "small" was not knowable in advance, which is the argument for the discipline rather than against it.

Still unaudited, because the inputs are not in this repo: the `hbm_weight` series (is the HBM revenue share a point-in-time estimate or a retrospective one?) and the 2018 McCallum$\rightarrow$TrendForce chain-link (is the splice ratio computed on overlap that runs past the observation date?). These remain the two highest Guard-4 risks in v2 and neither can be checked from what is published.

On the look-ahead penalty itself: fitting the confound regression on the full sample vs point-in-time moved the best forward F from 5.219 to 5.806 — the leak *deflated* the statistic here, where in the ag study the same correction cost half the result (36.0 $\rightarrow$ 17.8). The direction of the bias is not predictable. Point-in-time construction is a correctness requirement, not a haircut you can estimate and apply afterwards.

7. Guard 8 — a new one, added here

Check for definitional overlap before interpreting a lead.

Semiconductor industrial production (`IPG3344S`, NAICS 3344) against INDPRO produces `LEADS at 2m, p = 1.9e-08, F ratio 15.17` — the largest statistic in the entire run. It is also an accounting identity: IPG3344S is *a component of* INDPRO. Nothing was predicted.

The agricultural harness has no guard for this because its signals and targets came from different agencies measuring different things. In semiconductors the signal and the macro target are routinely built by the same statistical agency from overlapping source data, so the trap is live. It is now documented in the battery as the worked example.

8. The potash lesson, applied and closed out

Under administered pricing the market clears through quantity, so volume is the instrument — the recognition that turned an uninformative potash null into a real rejection. The DRAM analogue is semiconductor output. Non-circular targets:

Test	n	Granger verdict	Guard 5
semi IP → PAYEMS	653	LEADS at 2m ($p = 0.0018$)	**DIES** — 44.6% F retained, $p = 0.35$
semi IP → real GDP	217	LEADS at 1q ($p = 0.0057$)	**DIES** — 30.5% F retained, $p = 0.56$
semi IP deseasonalized (PIT) → PAYEMS	533	LEADS at 2m ($p = 0.0037$)	**DIES** — 70.7% F retained, $p = 0.11$

And the placebo that settles it: **total industrial production → PAYEMS leads at 1 month with p = 1.7e-20.** Generic output leads employment overwhelmingly, because employment is a lagging indicator. “Semiconductor output leads employment” was never a statement about semiconductors.

9. What v2 passes, kept and stated

Worth leading with, because it is genuinely uncommon:

1. **Reverse-direction testing** — v2 already reports GDP → R ($F = 2.14$, $p = 0.121$). Most lead-lag claims never run it.
2. **F-dominance** — 5.83 / 2.14 = **2.72**, above the 2.0 threshold. Report the ratio; it is a stronger and more honest statement than “the reverse was insignificant”, and it is immune to the underflow problem that produced the ag study’s third false positive.
3. **Lag-order robustness** across `p ∈ {1,2,3,4}` and Toda–Yamamoto.
4. **Having a placebo at all** — the ag study had none. The specific placebo needs replacing (§5), not the practice.

Not verifiable from this repository: the v2 Granger result itself. The paper’s Availability section promises “Granger causality estimation code (Stata and Python)” at this repo; `scripts/` contains one unrelated `.mjs` file, and the quarterly ASP series the VAR runs on is not here. The headline number cannot currently be reproduced by a reader. That is now fixable — the harness exists; it needs the data.

10. Sample size on the bifurcated index

`ram_commodity_R`, `ram_ai_R` and `hbm_weight` are populated for **three rows**: 2024, 2025, 2026 — one of which is the forecast row. No inferential statistic is estimable; the harness floor alone requires 5 observations for a single-lag test.

The honest label is **UNTESTABLE** — an absence of a test, not a null. The ag study retired its Bos indicus track on exactly this basis rather than reporting a rejection. v3 should either build the sub-indices quarterly (n $\approx$ 12, still marginal) or present the bifurcation as a descriptive regime claim with no inferential statistics attached, and make sure the composite's Granger result cannot be read as covering the sub-indices.

11. The theoretical payoff

The most useful non-statistical output of the ag work is a **mechanism with a boundary**:

> DRAM leads because memory is bought by a few hundred capacity-planning firms ahead of deploying compute, so the purchase *is* a forward expectation expressed as a price. Agricultural inputs are bought by ~2M atomized price-takers on an agronomic calendar, funded by last season's cash — no agent expresses a forward expectation as a price, so the failure is structural.

That converts RI from "DRAM happens to work" into a **class of indicator with a testable scope condition**: concentrated buyers + discretionary forward commitment + fast, elastic, private-market price. Agriculture becomes a tested-and-failed boundary case rather than an unexamined gap. Farm machinery — financed capex, the closest structural analogue to DRAM — leads nothing either, which sharpens the condition to *concentration* rather than merely *capex*.

The §5 and §8 results sharpen it further, at v2's expense: neither the broad semiconductor price complex nor semiconductor output satisfies it. Whatever the RI is measuring, it is narrower than "semiconductors", and v3 has to say so.

This is the strongest thing to import, and it costs no new data.

11a. RI-M: the monthly index, built and rejected

Run output: `analysis/out/FACTS-RIM.txt`. Code: `ri_leadtest/mccallum.py`, `ri_leadtest/index_v3.py`, `battery_v3.py`.

The archive is recovered. jcmit.net lapsed and the McCallum workbook is gone from the live web. It was retrieved from the Internet Archive — OLE metadata reads *Author: John McCallum, created 1998-11-09, last saved 2020-09-20* — and is cached at `analysis/data/mccallum_MemDiskPrice-xl95_20200920.xls`. 380 dated observations; 342 monthly $/MB values from 1985-01 to 2020-09.

RI-M was built from it and it works as a construction. 278 adjacent-month log returns, point-in-time by proof (`assert_point_in_time`: 193 overlapping months, max abs difference 0.0 when the archive is truncated and the index rebuilt), gap-aware (63 gaps totalling 150 months emit no return rather than a fabricated zero), vintage-stamped. It is everything v2's construction is not.

And it is not a usable index. Guard 6 and Guard 7 both fire:

Check	Result
Staleness	21.6% of months unchanged; longest flat run 10 months
vs IR213 import price index, monthly	r = **0.133**
vs IQ213 export price index, monthly	r = **0.029**
vs semiconductor PPI, monthly	r = **0.120**

The archive is a sequence of *individual advertised retail quotes* — BYTE and JDR magazine ads, later NewEgg listings — not a survey. The same quote is carried for months (`$504.833` runs from 1988-07 to 1989-04), and successive observations often switch to a different module. At annual frequency the noise averages out and it tracks everything at 0.45–0.60. At monthly frequency, which is the entire point, it agrees with nothing.

So RI-M's null is uninformative in exactly the sense Guard 6 was written for:

```
RI–M -> INDPRO  (n=278)  p = 0.622    F ratio 1.26
RI–M -> PAYEMS  (n=278)  p = 0.450    F ratio 17.54
RI–M -> real GDP (n=73)  p = 0.497    F ratio 1.82
```

Those are not marginal misses. Peak lead correlation is 0.122 and every one dies under Guard 5. But the instrument cannot support a rejection, so **the honest verdict is UNTESTABLE, not NO LEAD** — the same call the ag study made on cartel-administered potash prices, and for the same structural reason.

The conclusion this forces. The monthly Ramification Index cannot be built from public or recoverable sources. That is now established from two independent directions: no public *index* measures DRAM (§3, r = 0.186), and the one genuine *archive* of DRAM prices is too sparse and too stale to difference monthly. The binding constraint in §12 is confirmed, not relieved.

11b. Two data-integrity findings in the published series

The provenance in v2's README is wrong. v1 documents the series (`memory–index/research/ram–prices.md` §2) as "mid-year average retail for a typical mainstream module, **rounded**", blended from McCallum, hblok.net *and* TrendForce. v2's README describes it as "McCallum archive (1980–2017) chain-linked to TrendForce DDR4 contract ASP (2018–2026)". It is a rounded three-source blend, and v1 says so; v2 overstates it. This explains the r = 0.597 against the archive — that number is not evidence of an error, and I withdraw the stronger reading of it.

The 1988 RAM drought is missing from the index. This one is not a documentation issue.

Year	Published \$/MB	Published YoY	Archive \$/MB	Archive YoY
1987	133	−29.1%	158	−37.2%
1988	**107**	**−21.8%**	**361**	**+82.5%**
1989	73	−38.2%	300	−18.6%

v1's own §3 event table records 1988 as *"The Year of the RAM Drought … Prices spiked ~60% YoY"*, and the monthly archive shows it plainly: \$153/MB through mid-1987, \$199 by March 1988, \$505 from July 1988. The published price table shows a monotone decline straight through. **v1 §2 contradicts v1 §3 for that year**, and v2 inherits the price table unchanged.

This matters past 1988. v2's central new claim is RAMageddon — a supply-side divergence regime argued to be "without historical precedent" in the 46-year series. The nearest historical precedent is a post-dumping capacity-underinvestment shortage that the series does not contain. The uniqueness claim cannot be evaluated until 1988 is right.

Also: comparing v1's and v2's panels, **only the 2026 row changed** — by +86.9% (\$0.0052 → \$0.0097/MB). Every other year is byte-identical. The forecast row is live and moving between versions, which is the §6 point with a number attached.

12. What v3 needs, in order

1. **Acquire a monthly DRAM ASP series.** This is the binding constraint on everything else — §2 shows the power gain, §3 shows no public substitute works. TrendForce or Counterpoint contract+spot monthly, back as far as the licence allows. Everything downstream is built and waiting.
2. **Re-run the battery on it.** `analysis/battery.py` needs one new series and a section; the guards, tests and output format already exist.
3. **Publish the estimation code and the ASP series** (or a licensed derivative) so the headline Granger result is reproducible. Currently it is not.
4. **Correlate the vendors' ASP series against each other**, not just their correlations with GDP. That is Guard 7; §3 shows it is not a formality.
5. **Run Guard 5 on the actual v2 quarterly specification.** It passes on the annual panel at 98.9% F retained, which is a real and reportable result.
6. **Audit `hbm_weight` and the 2018 chain-link** for point-in-time construction. Not checkable from published data.
7. **Add a vintage column** to `indices-wide.csv` and mark 2026 provisional.
8. **Replace the NAND placebo** with one that fails the mechanism — passive components work, and are in the battery already.
9. **Downgrade the bifurcated index to descriptive**, or rebuild it quarterly.
10. **Write the scope condition into the theory section.**

Guards 1–3 v2 already passes, and leading with that is fair — provided 4–8 are run rather than asserted.

Files

`analysis/ri_leadtest/leadtest.py`	The harness. 8 guards, frequency-aware
`analysis/ri_leadtest/data.py`	Keyless FRED fetchers, cached to `analysis/data/`
`analysis/battery.py`	Every guard, every frequency
`analysis/out/FACTS.txt`	Verbatim run output. Every figure above traces here
`analysis/tests/`	20 regression tests pinning the guards